S. SADRA MOUSAVI

Tehran, Iran

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EDUCATION

Shahid Beheshti University Tehran, Iran — Sep 2022 – Present

B.Sc. in Computer Engineering

GPA: 18.29/20

Allameh Helli "NODET" High School Tehran, Iran — Sep 2019 – Aug 2022

High School Diploma, Mathematics and Physics

GPA: 19.39/20 (4/4)

RESEARCH INTERESTS

Embedded Systems
Edge Computing
Cybersecurity
Digital Twins
Embedded AI
Internet of Things (IoT)
Computer Networks

HONORS & AWARDS

Top 10% Academic Standing sep. 2025

Ranked among the top 10% of undergraduate students in the Computer Engineering cohort (approximately 100 students).

Iranian National University Entrance Examination sep. 2022

Achieved Rank 71 among more than 37,000 participants in the Regional Quota.

SKILLS

Programming: C/C++, Python, Java, C#

Low-Level Programming: x86 Assembly, ARM Assembly

Hardware Description Languages: Verilog

Embedded: STM32, ESP32, ARM Cortex-M, Arduino

Frameworks & Libraries: Scikit-learn, Pandas, NumPy, Matplotlib

Databases: PostgreSQL, MySQL

Tools: Git, Docker, Linux, MATLAB, Simulink

Soft Skills: Leadership, Teamwork, Communication, Time Management, Presentation

SELECTED PROJECTS

Model-Based Embedded Real-Time Heating Controller View Source Code

- Designed and modeled a real-time multi-room heating control system using MATLAB Simulink and Stateflow following the V-Model development process.
- Implemented thermostat logic, safety checker, and room temperature dynamics using model-based design and requirement-driven architecture.
- Performed simulation, validation, and test coverage analysis using Simulink Test and Embedded Coder workflows.
- Technologies: MATLAB, Simulink, Stateflow, Model-Based Design, C

Operating Systems Algorithms and Memory Management View Source Code

- Implemented core operating system concepts including page replacement, virtual memory, synchronization, and process scheduling algorithms.
- Simulated and analyzed memory management techniques to evaluate their correctness and performance.
- Technologies: Python, Operating Systems, Virtual Memory, Process Scheduling.

Secure File Encryption System View Source Code

- Developed a secure file encryption application implementing classical and modern cryptographic algorithms.
- Implemented encryption, decryption, integrity verification, and secure key management techniques.
- Technologies: Python, Cryptography, File Security.

TEACHING EXPERIENCE

Teaching Assistant — Advanced Programming Fall 2023 – Spring 2025
Dr. Mojtaba Vahidi-Asl

- Served as Head of the Workshop Team, coordinating teaching assistants responsible for workshop sessions.
- Delivered programming workshops and presented advanced programming concepts to undergraduate students.
- Contributed to workshop planning and instructional activities across four consecutive semesters.

Teaching Assistant — Operating Systems Concepts Fall 2024 – Spring 2025
Prof. Mohsen Ebrahimi Moghadam

- Designed programming assignments and course projects covering operating system concepts.
- Assisted in presenting course materials and guiding students throughout project implementation.
- Supported students in understanding operating systems through discussions and technical guidance.

Teaching Assistant — Computer Architecture Fall 2024 – Spring 2025
Dr. Dara Rahmati

- Designed programming assignments and course projects related to computer architecture.
- Assisted students in understanding processor architecture and low-level system concepts.

ACADEMIC SERVICE

Editorial Team Member — Faculty Scientific Magazine Spring 2024 – spring 2025
Supervised by Prof. Shams Aliee

- Contributed to the editorial and publication process of the faculty's scientific magazine.
- Assisted in reviewing, editing, and preparing academic content for publication.

Executive Committee Member — University Conference on Artificial Intelligence Fall 2024 – Spring 2025
Supervised by Dr. Salimi Badr

- Served as a member of the executive committee for the university's Artificial Intelligence Conference.
- Assisted in conference organization, coordination, and event execution.